

File Handling & Modules

1. File Handling Basics

File handling is an important part of any web application. Python has several functions for creating, reading, updating, and deleting files.

The key function for working with files in Python is the **open()** function. It takes two parameters; filename, and mode.

- "r" - Read (Default). Opens a file for reading, error if it doesn't exist.
- "a" - Append. Opens a file for appending, creates the file if it does not exist.
- "w" - Write. Opens a file for writing, creates the file if it does not exist. (Overwrites existing content)
- "x" - Create. Creates the specified file, returns an error if the file exists.

2. The with Statement

It is good practice to use the **with** keyword when dealing with file objects. The advantage is that the file is properly closed after its suite finishes, even if an exception is raised at some point.

Example:

```
with open("file.txt", "r") as f:  
    content = f.read()
```

3. CSV Handling

CSV (Comma Separated Values) format is the most common import and export format for spreadsheets and databases. Python has a built-in module called **csv** to handle these files.

You can use `csv.reader()` for reading and `csv.writer()` for writing data to CSV files.

4. OS and Sys Modules

- **os module:** Provides a portable way of using operating system dependent functionality. (e.g., `os.path.exists()`, `os.remove()`, `os.mkdir()`)
- **sys module:** Provides access to some variables used or maintained by the interpreter and to functions that interact strongly with the interpreter. (e.g., `sys.argv` for command line arguments).

5. Modules and PIP

Consider a module to be the same as a code library. A file containing a set of functions you want to include in your application.

To create a module just save the code you want in a file with the file extension `.py`. To use a module, use the **import** statement.

PIP: PIP is a package manager for Python packages, or modules if you like. You can use it to install packages from the Python Package Index (PyPI).